NEWSLETTER

FEBRUARY 2025

WHY COOL THE BATTERY?!

THE DILEMMA OF PERFORMANCE AND COMPLEXITY

BALANCING THE BRAKES

THE GEOMETRIC SOLUTION FOR BRAKE SYSTEMS

WHAT CAN YOU EXPECT?

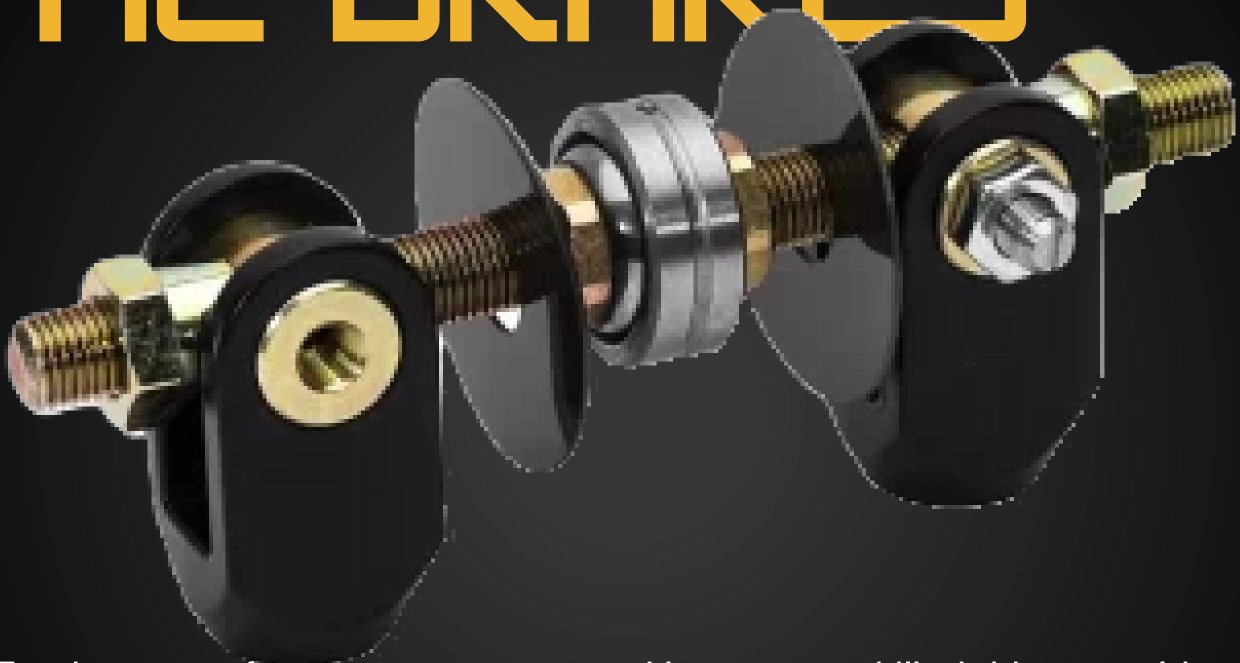
WHAT IS AUS RACING

AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

BALANCING THE BRAKES



For the most efficient deceleration, all tires need to slow down and stop at the same time.

Cars which are heavier towards the front generally have larger brakes in the front as they require more force to decelerate at the same rate as the rear brakes.

Commercial cars are generally biased forward. In case of an emergency, if the driver applies a hard brake, causing the front

wheels to lose grip and lock up first, the car won't spin out, making it much safer.

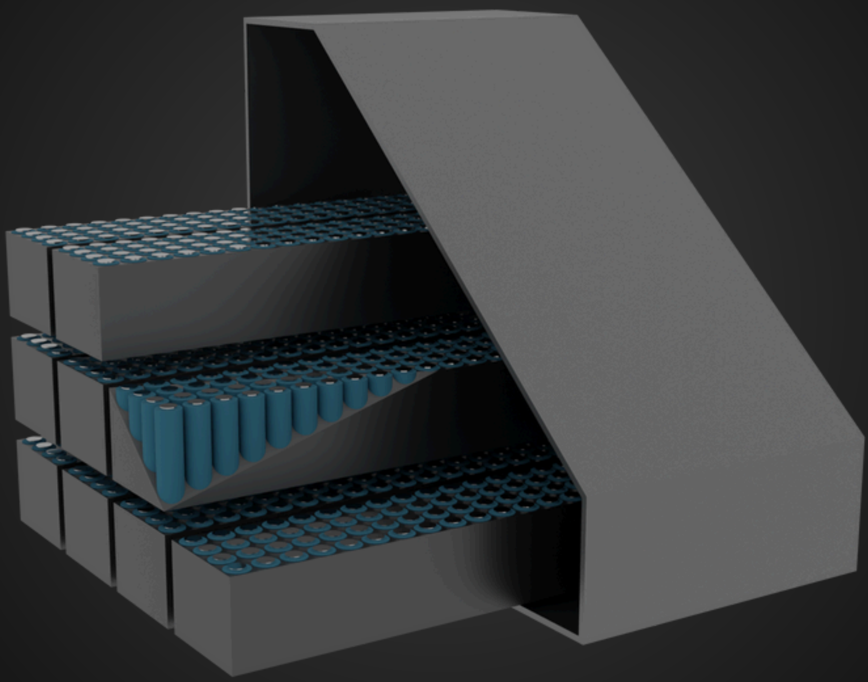
However, a skilled driver would prefer a rearward bias to turn the car faster.

Therefore, the brake bias needs to be adjustable to get the most performance out of each driver as they all have different preferences.

This can be achieved with a balance bar. A balance bar is a part mounted at the brake pedal. Based on it's angle, the force applied gets distributed between the front and rear brakes.

The adjustable angle of the balance bar determines the bias.

COOKING THE BATTERY



The battery is a critical system of the car which is susceptible to overheating because of the low voltage and high current drawn by the motor.

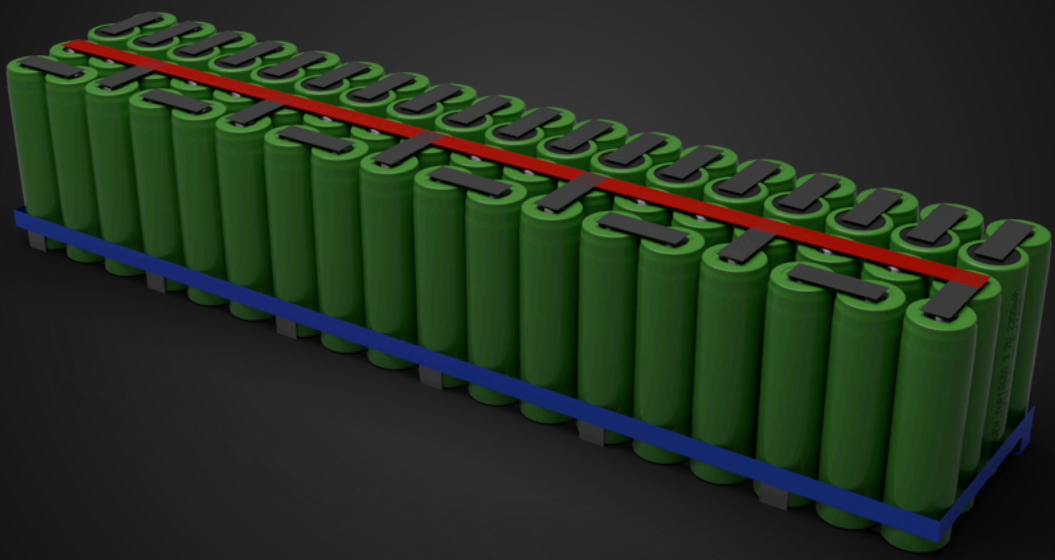
The heat generated by the battery cannot be dissipated passively, inevitably overheating the battery.

Lithium ion cells are extremely volatile. High temperatures can be hazardous and lead to catastrophic failure.

The battery can reach critical temperatures within a few laps, hence, making a cooling system a crucial part of completing the endurance race.

Air cooling is generally the easiest and most straightforward way to cool cells. However, a water cooled system is a lot more effective.

CONNECTING THE CELLS



The cells need to be connected in the predetermined electrical and mechanical configuration to get the ideal voltage and power output.

The mechanical configuration is important for saving space and getting the right airflow for cooling into the battery.

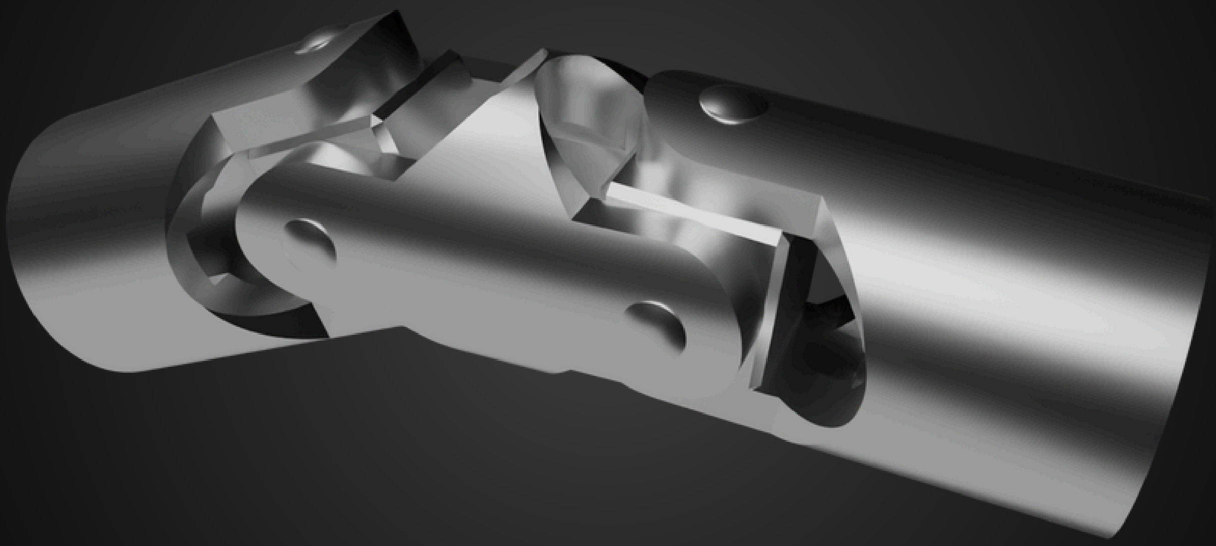
Wires could be connected to the cells, however, busbars are used.

Busbars have the same function as wires, conducting electricity.

The busbars and cell mounts need to be designed meticulously to ensure the battery has good electrical and mechanical integrity.

They also determine the resistance and add to the cooling load for the battery cooling system.

MOUNTING THE STEERING



Positioning of the steering system in the car contributes to driver comfort, drivability of the car, and structural integrity.

The steering wheel itself needs to be placed in a position for allowing the driver to be as comfortable as possible. A comfortable driver will always perform better on the track.

The mounting location also affects the drivability of the car.

Primarily, due to the universal joints used. These joints have a property of being out of phase when at an angle less than 180°.

This means, rate of change of the angle at the steering wheel is out of phase from the rate of change of the angle at the wheels.

This phase difference can be jarring and make the car difficult to control. Therefore, ideally, a larger angle needs to be used.

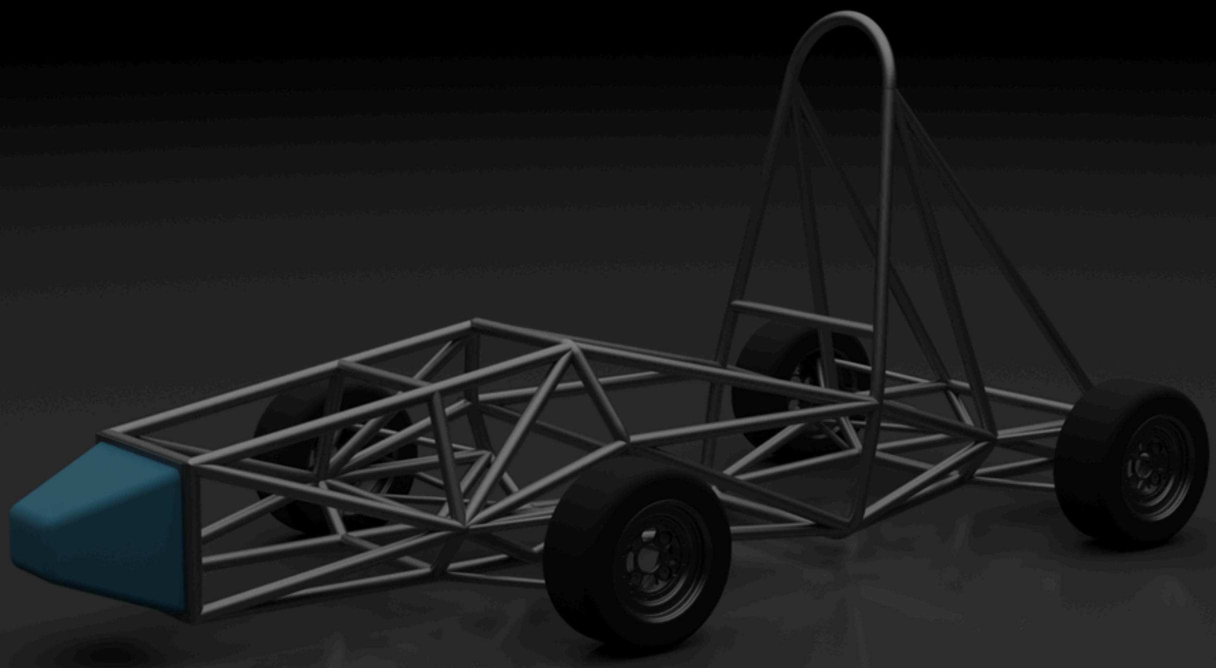
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact oaaa@aus.edu to learn more about getting involved with the AUS Racing team.



AUS

American University of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH GROUP

ausracingfsae https://ausracing.netlify.app/ ausracing@aus.edu